

Finance Quiz/Tutor Bot

Course Name: DevOps Foundation

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

In today's digital era, financial literacy is becoming increasingly important, yet many individuals struggle to understand financial concepts due to lack of personalized learning resources. Traditional learning platforms provide static content, which often fails to engage users or adapt to their individual learning pace.

Financial education systems usually lack:

- Interactive learning mechanisms
- Real-time feedback
- Personalized guidance

As a result, users find it difficult to retain knowledge and stay motivated.

To address these challenges, there is a need for an intelligent system that can dynamically generate quizzes, evaluate user responses, and provide personalized explanations. **Generative AI** can solve this problem by enabling adaptive, interactive, and user-specific learning experiences.

1.2 Project Objectives

The main objective of the **Finance Quiz & Tutor Bot using Generative AI** project is to provide an interactive and personalized financial learning platform.

Key objectives include:

- **To develop an AI-based quiz generation system**

The system dynamically generates finance-related quizzes based on user input and difficulty level.

- **To provide real-time evaluation and feedback**

Users receive instant scoring and AI-generated explanations for their answers.

- **To enhance financial learning through personalization**

The bot adapts content according to user performance and understanding.

- **To improve engagement using conversational AI**

The tutor bot interacts with users like a virtual mentor.

- **To make financial education accessible and interactive**

The system ensures easy access and engaging learning for all users.

1.3 Scope of the Project

The scope of this project is to design and develop a **Generative AI-based Finance Quiz and Tutor Bot** that enhances financial learning through automation and intelligence.

The project focuses on:

- Dynamic quiz generation using AI
- Answer evaluation and explanation
- Interactive chatbot-based tutoring
- User-friendly interface for learning

The system allows users to:

- Select topics and difficulty levels
- Attempt quizzes
- Receive instant feedback

However, this project focuses on **educational functionality** and does not include real-time financial transactions or full-scale financial services.

2. Proposed Solution

2.1 Key Features

The system includes the following key features:

- AI-powered quiz generation
- Real-time answer evaluation
- Personalized explanations
- Interactive tutor chatbot
- User-friendly interface

- Adaptive difficulty levels
 - Faster and engaging learning experience
-

2.2 Overall Architecture / Workflow

The system follows an intelligent workflow for delivering financial education:

1. User Interaction

Users access the platform and select a financial topic and difficulty level.

2. Input Processing

The system processes user input and sends it to the Generative AI model.

3. Quiz Generation

AI dynamically generates quiz questions based on selected parameters.

4. User Response

Users attempt the quiz and submit their answers.

5. Evaluation

The system evaluates responses using AI and predefined logic.

6. Explanation Generation

AI provides detailed explanations for correct and incorrect answers.

7. Tutor Bot Interaction

Users can interact with the chatbot to clarify doubts.

8. Result Display

Scores and feedback are displayed to the user.

9. Continuous Learning

The system adapts future quizzes based on user performance.

Fig 2.2 Overall Architecture

(User → Frontend → Backend → Generative AI Model → Response → User)

Fig 2.2 Workflow Diagram

(Input → AI Processing → Quiz → Evaluation → Feedback → Learning Loop)

This workflow ensures **personalization, automation, and engagement.**

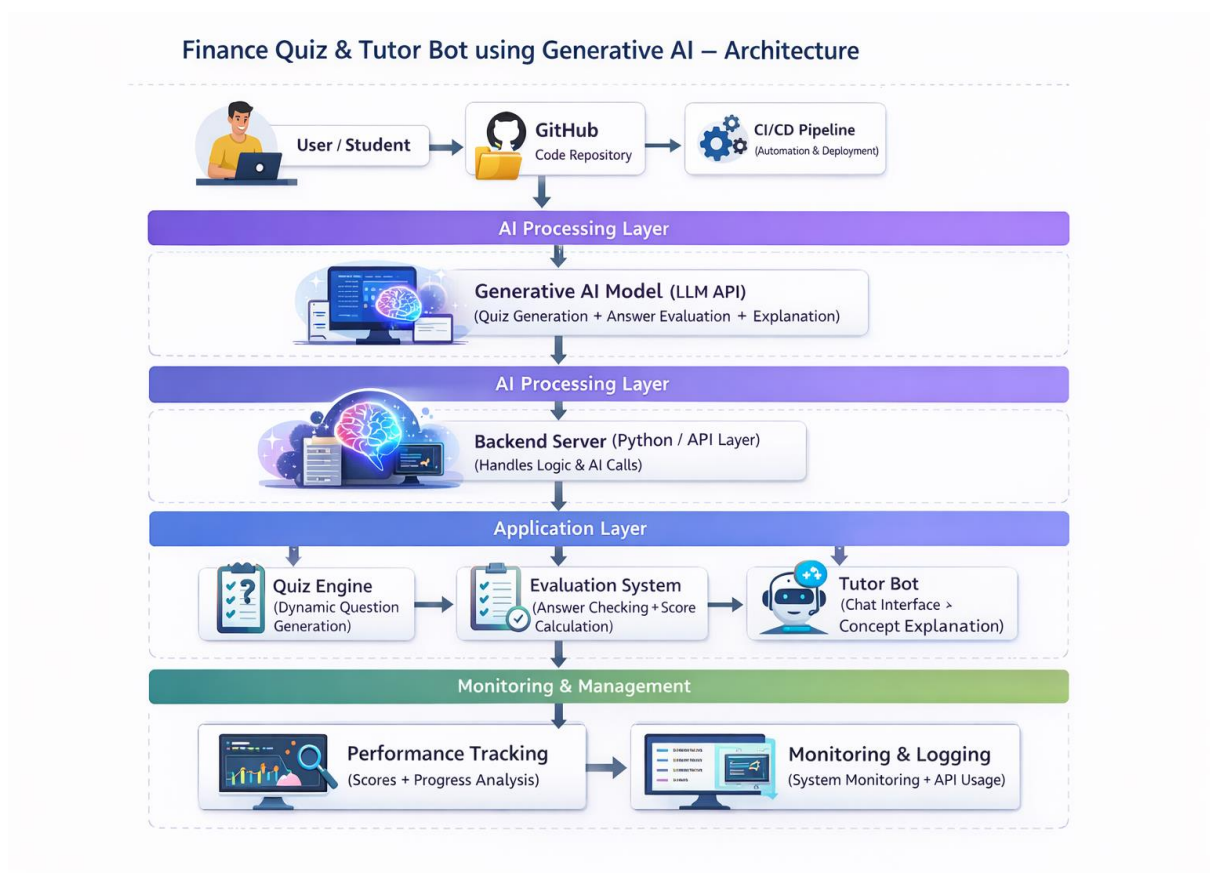


Fig 2.2 Over all Architecture

AI-Powered Finance Quiz & Tutor Bot Workflow

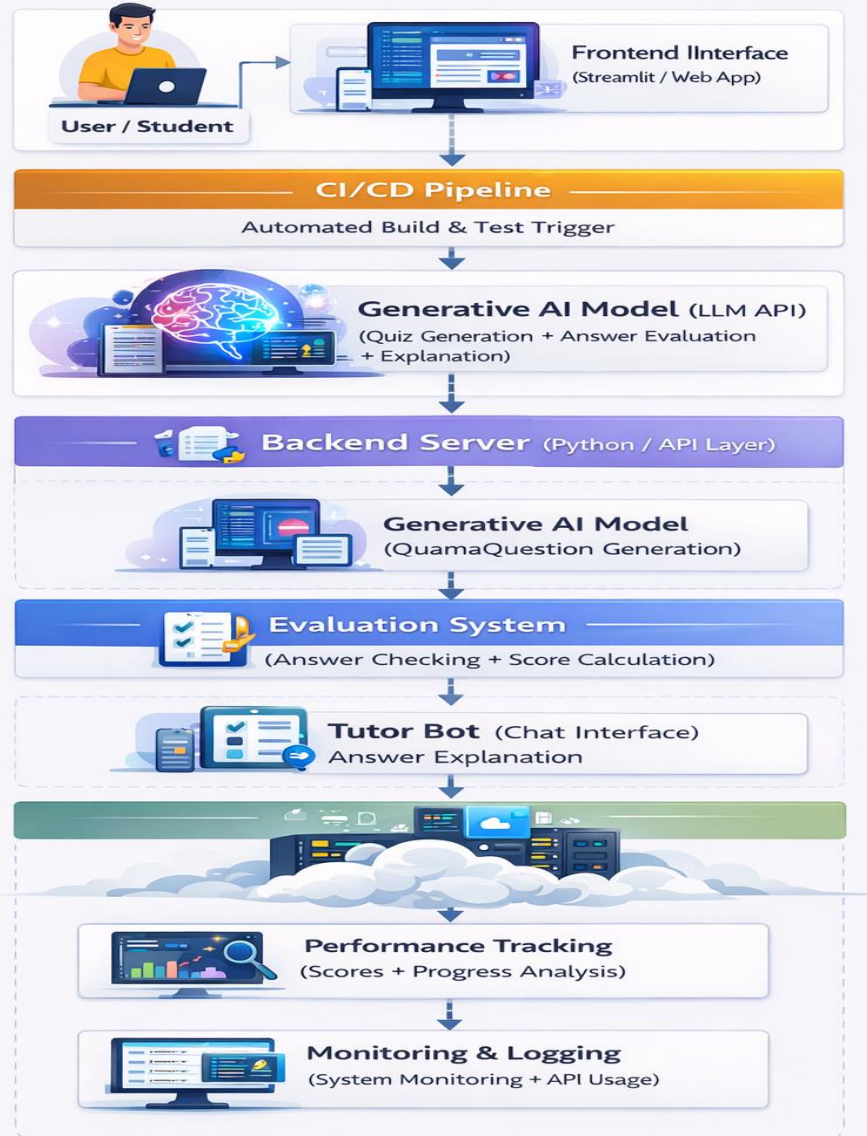


Fig.2.2 Workflow Diagram

This workflow ensures **automation, reliability, and scalability in infrastructure management.**

2.3 Tools & Technologies Used

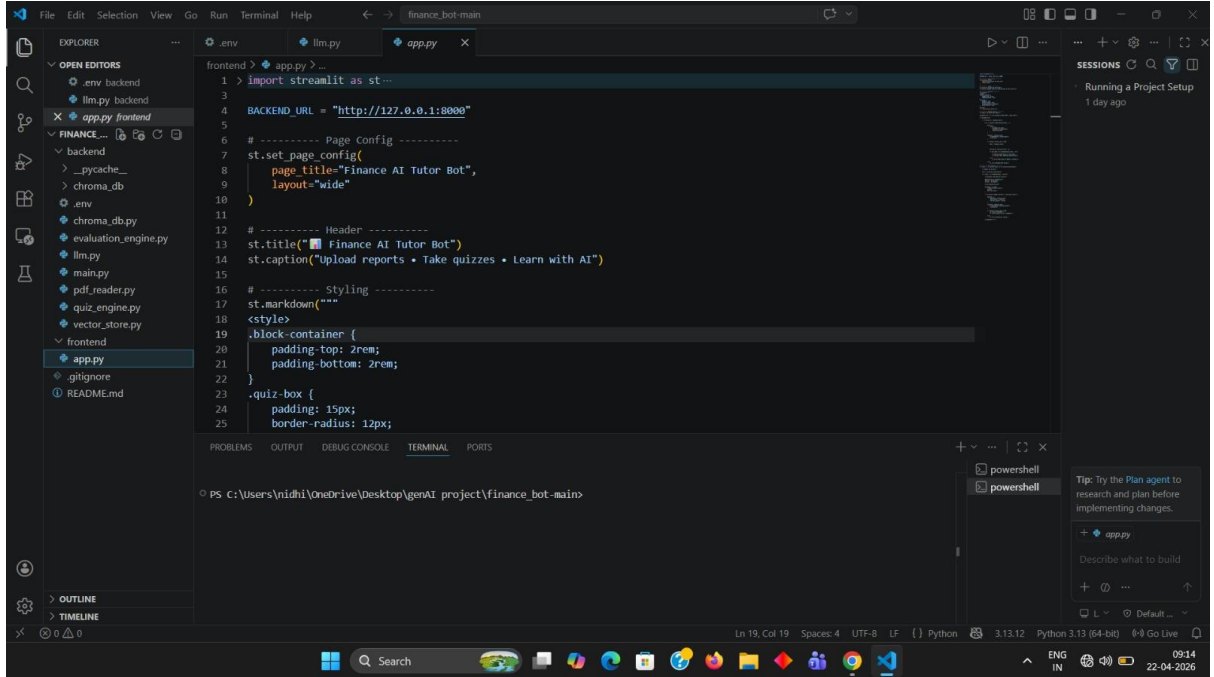
Tool / Technology	Purpose
Generative AI (LLM)	Quiz generation & explanations
Python	Backend development
Streamlit	User interface
Git	Version control
GitHub	Code repository
JavaScript	Frontend interactivity
HTML, CSS	UI design
APIs	AI model integration, Groq API
Cloud Environment	Hosting application

3. Results & Output

3.1 Screenshots / Outputs

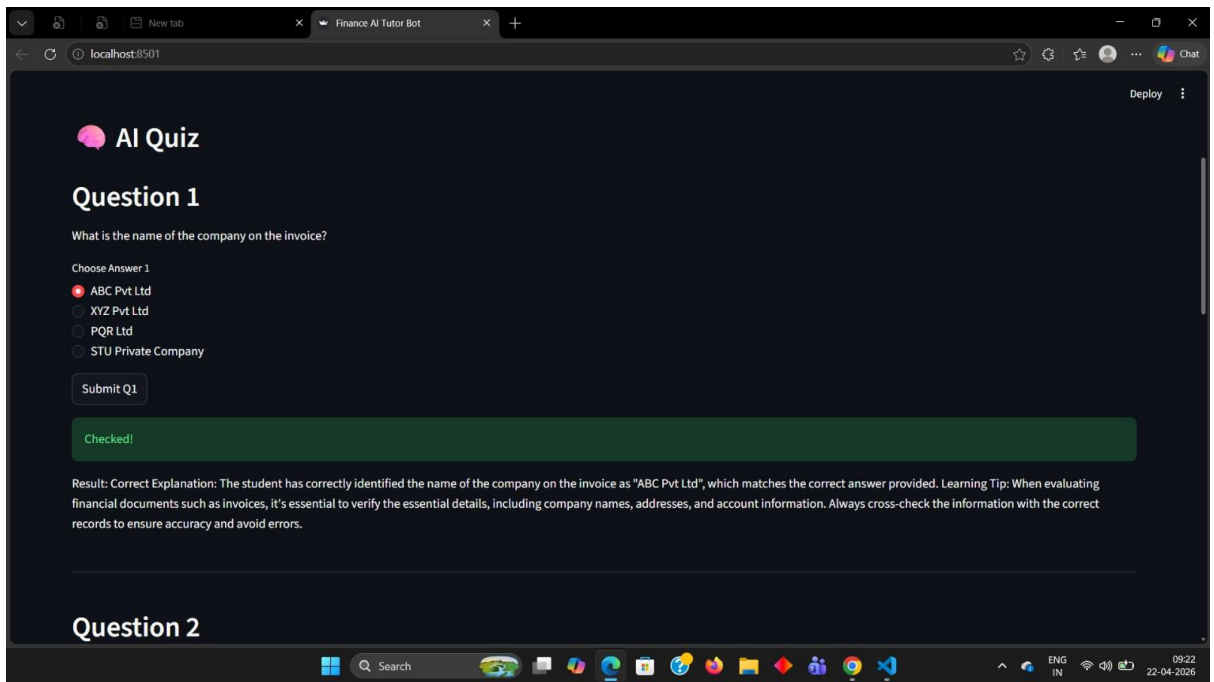
The project produced several outputs demonstrating successful implementation:

- Quiz generation interface
- AI-generated questions
- Answer evaluation results
- Chatbot interaction screen
- Final score display



```

1 > import streamlit as st
2
3
4 BACKEND_URL = "http://127.0.0.1:8000"
5
6 # ----- Page Config -----
7 st.set_page_config(
8     page_title="Finance AI Tutor Bot",
9     layout="wide"
10 )
11
12 # ----- Header -----
13 st.title("🧠 Finance AI Tutor Bot")
14 st.caption("Upload reports • Take quizzes • Learn with AI")
15
16 # ----- Styling -----
17 st.markdown("""
18 <style>
19 .block-container {
20     padding-top: 2rem;
21     padding-bottom: 2rem;
22 }
23 .quiz-box {
24     padding: 15px;
25     border-radius: 12px;
  
```



AI Quiz

Question 1

What is the name of the company on the invoice?

Choose Answer 1

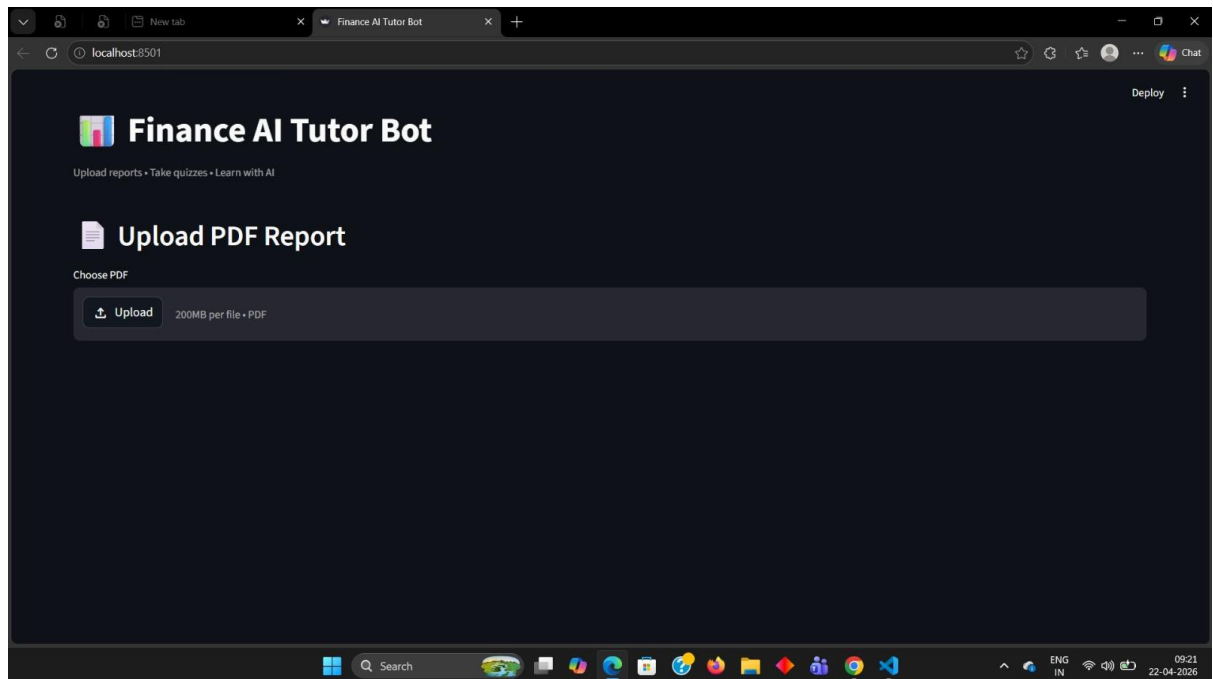
- ☒ ABC Pvt Ltd
- ☐ XYZ Pvt Ltd
- ☐ PQR Ltd
- ☐ STU Private Company

Submit Q1

Checked!

Result: Correct Explanation: The student has correctly identified the name of the company on the invoice as "ABC Pvt Ltd", which matches the correct answer provided. Learning Tip: When evaluating financial documents such as invoices, it's essential to verify the essential details, including company names, addresses, and account information. Always cross-check the information with the correct records to ensure accuracy and avoid errors.

Question 2



3.2 Reports / Dashboards / Models

The system generates:

- Quiz results and performance reports
- AI-generated explanations
- System workflow diagrams
- Architecture diagrams

These models help in understanding system behavior and performance.

3.3 Key Outcomes

Successful implementation of AI-based quiz system

The system dynamically generates quizzes using Generative AI.

- **Improved learning engagement**

Interactive quizzes and chatbot increase user interest.

- **Personalized learning experience**

Content adapts according to user performance.

- **Instant feedback and explanation**

Users understand concepts immediately.

- **Efficient and scalable solution**

System can handle multiple users and topics.

4. Conclusion

The **Finance Quiz & Tutor Bot using Generative AI** project successfully demonstrates how AI can transform traditional learning into an interactive and personalized experience.

By integrating Generative AI with modern web technologies, the system enables:

- Dynamic quiz generation
- Real-time evaluation
- Intelligent tutoring

This approach improves learning efficiency, engagement, and accessibility.

The project highlights the importance of **AI in education**, especially in making complex subjects like finance easier to understand.

Overall, the system provides a scalable and innovative solution for modern financial education.

5. Future Scope & Enhancements

The project can be further improved by:

- Integrating voice-based interaction (AI voice assistant)
- Adding multilingual support
- Using advanced models for better accuracy
- Tracking user progress with analytics dashboard
- Implementing gamification (badges, rewards)
- Deploying on mobile applications
- Integrating real-world financial datasets

These enhancements will make the system more powerful, interactive, and suitable for real-world applications.